Pb Action

Lead Material-Flow Model

Method Overview & Technical README

A transparent, single-pass (back-of-the-envelope) accounting of lead flows through a country's lead-acid battery economy, anchored on reported refined-lead production.

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This is a first-order *screening* model built on publicly available trade and production data with literature-derived efficiency assumptions. It is a single deterministic pass, *not* a calibrated fit: its outputs are best read as indicative approximations and structural diagnostics, not precise tonnages. Its purpose is to show, transparently and for any country, roughly where lead goes and which stage of the recycling loop looks weak. Where calibration, a formal/informal split, or quantified uncertainty are needed, the companion *mass-balance* model (documented separately) is the stronger tool.

Contents

1 Purpose and scope

This document describes a country-level material-flow model of lead moving through a lead-acid battery (LAB) economy. Its purpose is **rapid, transparent screening**: for any country with a reported refined-lead figure, it estimates the size of the recycling loop — how much end-of-life battery lead is generated and collected, how much scrap is smelted, and how much lead is manufactured into new batteries — and shows which stage of that loop appears under- or over-supplied. It is *not* intended to produce precise tonnages. Its outputs are approximations and structural diagnostics rather than point figures.

The model traces lead around a single loop and **anchors the whole picture on one measured quantity: a country's reported refined-lead (smelting) output**. From that anchor it solves in two directions. A **backward chain** works upstream from *secondary* (recycled) smelting to estimate the collected and generated end-of-life batteries that must have fed it. A **forward chain** works downstream from *total* refining to estimate the lead manufactured into new batteries and installed. Trade adjusts each pool along the way.

All quantities are expressed in **tonnes of contained lead per year**. Trade data are drawn from BACI bilateral statistics; mine and refined production from the British Geological Survey (BGS) and the US Geological Survey (USGS). Because a single year of customs data is noisy, the model is normally read over a three-year average. It carries **no time dynamics**: one year's flows are treated as a steady-state snapshot of the whole loop, with no lag between a battery being installed and later retiring.

1.1 Relationship to comparable methodologies

This model belongs to the established tradition of material-flow analysis (MFA) for lead and draws on that literature for its efficiency priors. Dente et al. (2025) reconstruct in-use lead stocks across eleven countries and provide independent in-use-stock benchmarks; Dong et al. (2025) present a detailed dynamic China MFA; and Tür et al. (2016) develop field methods for estimating used-lead-acid-battery volumes in African settings. Relative to those full dynamic analyses, the present model is deliberately lighter: a single deterministic pass with no stock reconstruction, no uncertainty propagation, and no formal/informal disaggregation. That lightness is the point — it runs instantly for any country from public data — but it also bounds what the model can claim (Sections 4 and 5), and it is why a calibrated companion model exists for questions this one cannot answer.

2 The processing chain

2.1 Structure

Lead moves through the loop shown in Figure 1 (final page). Two production stages run forward — **primary smelting** of ore and **manufacturing** of refined lead into batteries — and two recycling stages run in reverse of the physical flow when the model solves them — **breaking** of spent batteries and **secondary smelting** of the resulting scrap. The pools between these stages are: an *ore* pool, a *scrap* pool, a *feedstock* (refined-lead) pool, and a *battery* pool. Collection feeds spent batteries into the loop; installation is where manufactured and imported batteries enter service.

DECISION

The anchor is reported refined-lead output. Rather than build tonnages up from an assumed collection rate (which would compound every assumption), the model pins the chain to a measured quantity — reported refined-lead production — and solves outward. The recycling chain is anchored specifically on the *secondary* (recycled) portion of that output, so the estimated collection and breaking volumes are whatever is required to have produced the recycled lead a country actually reports.

DECISION

Splitting reported refining into primary and secondary. Only the secondary portion anchors recycling, so total refined output must be divided. Where the production source reports the split directly (USGS), it is used as reported. Where only a single total is available (BGS), the model estimates primary output from domestically available ore and treats the remainder as secondary; if that ore-based estimate is below 5% of total refined output, the country is assumed to run no meaningful primary smelter and *all* of its refined lead is treated as secondary (Equation 2).

2.2 Symbols

Trade enters as annual-average lead content. For each traded commodity c, let I_c be imports and E_c exports (tonnes Pb/yr); a *net* flow is $I_c - E_c$. The commodities are ore, scrap, used batteries, crude lead, refined feed, and finished batteries. Production inputs are mine output M and reported refined lead R, the latter split into primary R_{pri} and secondary R_{sec}. The process parameters (γ, δ, β, and the recovery efficiencies η) are defined in Table 2.

2.3 Equations

Anchor and split. Domestically available ore gives an estimate of primary metal; the balance of reported refining is secondary:

$$O = M + I_{\text{ore}} - E_{\text{ore}}, \qquad \text{(ore available domestically)} \tag{1}$$

$$R_{\text{pri}} = \begin{cases} 0, & \text{if } \eta_{\text{ore}} \max(0, O) < 0.05\,R \\ \eta_{\text{ore}} \max(0, O), & \text{otherwise} \end{cases} \qquad \text{(primary refined)} \tag{2}$$

$$R_{\text{sec}} = R - R_{\text{pri}}. \qquad \text{(secondary refined — the recycling anchor)} \tag{3}$$

Backward chain (recycling). Starting from secondary smelting output, each step inverts one transformation to recover the quantity that must have entered it. Net trade of an intermediate is removed where it was added downstream (imports subtracted, exports added back), so the result is the *domestically originated* flow:

$$Q_{\text{scr}} = R_{\text{sec}} \,/\, \eta_{\text{sec}}, \qquad \text{(scrap the secondary smelter consumed)} \tag{4}$$

$$Q_{\text{brk}} = \max(0,\ Q_{\text{scr}} - I_{\text{scr}} + E_{\text{scr}}), \qquad \text{(lead broken out of batteries locally)} \tag{5}$$

$$Q_{\text{col}} = Q_{\text{brk}} \,/\, (\delta\, \eta_{\text{brk}}), \qquad \text{(lead present in collected batteries)} \tag{6}$$

$$C = \max(0,\ Q_{\text{col}} - I_{\text{use}} + E_{\text{use}}), \qquad \text{(domestically collected batteries)} \tag{7}$$

$$G = C \,/\, \gamma, \qquad \text{(total end-of-life batteries generated)} \tag{8}$$

$$W = (1 - \gamma)\, G. \qquad \text{(uncollected / permanently disposed)} \tag{9}$$

Forward chain (production). Starting from total refined lead $R = R_{\text{pri}} + R_{\text{sec}}$, the feedstock pool is assembled, split between battery and non-battery uses, and carried through manufacturing to installed batteries. Net crude-lead trade is refined into the pool (with a recovery loss); net refined-feed trade joins it directly:

$$\Phi = R + (I_{\text{fd}} - E_{\text{fd}}) + \eta_{\text{ref}}\,(I_{\text{cr}} - E_{\text{cr}}), \quad \text{(refined-lead feedstock pool)} \tag{10}$$
$$N = (1-\beta)\,\Phi, \quad \text{(lead to non-battery uses)} \tag{11}$$
$$P = \eta_{\text{mfg}}\,\max(0,\,\beta\,\Phi), \quad \text{(lead in new batteries — manufacturing)} \tag{12}$$
$$A = P + I_{\text{bt}} - E_{\text{bt}}. \quad \text{(implied installation — apparent consumption)} \tag{13}$$

Every quantity is computed in one pass, in the order above; nothing is solved iteratively. A reader with the trade totals, the reported refined figure, and the parameter values can reproduce all of it by hand.

ILLUSTRATIVE CALCULATION

Take a country reporting $R = 100{,}000$ t of refined lead, all secondary ($R_{\text{sec}} = 100{,}000$, $R_{\text{pri}} = 0$), with every trade flow set to zero for simplicity, at the default parameters of Table 2.
Backward: $Q_{\text{scr}} = 100{,}000/0.97 = 103{,}093$; $Q_{\text{brk}} = 103{,}093$ (no scrap trade); $Q_{\text{col}} = 103{,}093/(0.95 \times 0.95) = 114{,}232$; $C = 114{,}232$; $G = 114{,}232/0.95 = 120{,}244$; $W = 0.05 \times 120{,}244 = 6{,}012$.
Forward: $\Phi = 100{,}000$; $N = 0.15 \times 100{,}000 = 15{,}000$; $P = 0.98 \times (0.85 \times 100{,}000) = 83{,}300$; $A = 83{,}300$.
Reading: to have produced 100 kt of recycled refined lead, the country must have generated about 120 kt of end-of-life battery lead (of which ~6 kt went uncollected) and manufactured about 83 kt of lead into new batteries.

2.4 Product categories and lead-content conversion factors

All flows enter the model in tonnes of contained lead. Customs records report gross product mass, which is converted to lead content with a product-specific factor; Table 1 gives the factor for each HS code the model uses. In practice these fractions vary by product grade and region, so they are approximations and are adjustable inputs, not fixed constants.

HS	Product	Factor	Range
260700	Ore / concentrates	0.60	0.45–0.75
780110	Refined unwrought lead	0.99	fixed
780191	Antimonial lead	0.96	0.94–0.98
780199	Other unwrought (crude) lead	0.97	0.95–0.99
780200	Lead scrap	0.725	0.50–0.95
282410	Lead monoxide	0.93	fixed
282490	Other lead oxides	0.90	0.87–0.93
850710	SLI (starter) batteries	0.60	0.55–0.65
850720	Other lead-acid batteries	0.60	0.55–0.65
850790	Battery parts	0.625	0.30–0.95
854810	Used / waste batteries	0.60	0.55–0.65

Table 1: Lead-content conversion factors by HS product category. Factors convert gross traded mass to contained lead; all model flows are in lead-content tonnes. Any code not listed falls back to a 0.70 fraction.

2.5 Treatment of specific trade codes

DECISION

Where each traded product attaches to the chain. Scrap (HS 780200) attaches at the secondary smelter (Equation 5). Used batteries (HS 854810, or its later code 854911) attach at collection (Equation 7). Crude unwrought lead (HS 780199) is refined into the feedstock pool with a recovery loss η_{ref}, whereas refined lead and lead oxides (HS 780110/780191/282410/282490) join the pool directly (Equation 10). Finished batteries (HS 850710/850720/850790) do not pass through manufacturing — they add directly to installation (Equation 13). Non-battery lead articles are treated as leaving the pool with the non-battery branch (Equation 11).

An alternative anchor is available where a country reports measured battery *collection* rather than relying on smelting: the backward chain can instead start from that collected tonnage (converted to lead content) and run forward through breaking and smelting. The smelting anchor is the default because refined-production data are available for far more countries.

3 Parameters

Table 2 lists the model's parameters, their defaults, and the basis for each. Every parameter is an efficiency or share in $[0, 1]$ and is adjustable; the defaults are global literature midpoints. The single parameter that most rewards country-specific attention is the collection rate γ, which varies widely with a country's waste-management maturity.

Parameter	**Symbol**	**Default**	**Basis / status**
Collection rate	γ	0.95	Fraction of end-of-life battery lead collected; the priority country-specific lever (≈0.99 high-income, ≈0.60–0.70 low-income)
Breaking recovery	η_{brk}	0.95	Lead recovered when batteries are broken and separated
Secondary smelting recovery	η_{sec}	0.97	Net yield of scrap to refined lead
End-of-life Pb retention	δ	0.95	Lead still present in a battery at collection
Battery share of demand	β	0.85	Share of refined feedstock going to batteries vs. all other uses
Manufacturing efficiency	η_{mfg}	0.98	Lead retained through battery manufacturing
Primary smelting recovery	η_{ore}	0.95	Yield of ore/concentrate to primary metal
Refining recovery on crude	η_{ref}	0.97	Applied to net crude-lead trade entering the feedstock pool

Table 2: Model parameters and their default values. All are dimensionless recoveries or shares; all are adjustable, with the defaults set to literature midpoints.

4 What the model can and cannot tell you

The model's reliable outputs are *structural and comparative*. Anchoring on a measured quantity and applying a short, explicit list of recoveries, it answers questions such as: how large is a country's recycling loop relative to its battery manufacturing; which stage (collection, breaking, smelting, manufacturing) is the bottleneck; and whether a country is a net importer or exporter of recycling capacity. Because it runs identically for every country, it supports direct cross-country comparison and global mapping. Because every step is a single multiplication or division by a stated parameter, the whole calculation is auditable by hand and its sensitivity to any assumption is transparent.

What it *cannot* do follows from the same simplicity. It does not produce precise tonnages; it does not separate formal from informal processing; it does not represent the time lag between installation and retirement; and it inherits, without correction, any error in the reported refining anchor or the trade statistics. It does not reconcile its two chains against an independent constraint — so it can report an internally implausible picture without flagging it. For those tasks the calibrated companion mass-balance model is the appropriate tool: it adds a reconstructed in-use stock, a lifespan-driven retirement lag, parallel formal/informal lanes, and Monte-Carlo uncertainty, and it solves against explicit anchors so that inconsistencies surface rather than pass through silently.

5 Known limitations and open items

The model's limitations are deliberate consequences of keeping it a transparent, single-pass screen. Each is stated below alongside how the companion mass-balance model addresses it.

Single pass, not reconciled. The two chains are computed independently from the anchor and never checked against each other or against an external quantity. *The mass-balance model* instead solves against two anchors — reported secondary refining and a stock-derived installation target — so that any tension between reported inputs is localized and reported rather than hidden.

No formal/informal split. Collection, breaking, and smelting are single aggregate flows. *The mass-balance model* runs parallel formal and informal lanes with recovery efficiencies that differ sharply between them — the feature that most distinguishes lead economies in low- and middle-income countries.

No time dynamics. A single year is treated as a steady-state snapshot; the lag between a battery being installed and later retiring is absent. *The mass-balance model* carries an in-use stock and a growth-corrected retirement rate driven by an effective battery lifetime, so today's scrap reflects batteries installed years earlier.

No uncertainty quantification. Outputs are single point estimates. *The mass-balance model* propagates parameter ranges through a Monte-Carlo ensemble and reports fan bands, sensitivities, and the subset of draws consistent with the anchors.

OPEN

Anchor and split sensitivity. Results move with the choice of production source (BGS vs. USGS), and the "5%-of-refined" rule that classifies output as all-secondary can misjudge small primary producers. Where the two sources disagree materially, the recycled-lead-measuring source is generally the safer anchor.

OPEN

Global default parameters. Every recovery is a literature midpoint rather than a country-measured value; only the collection rate γ is routinely adjusted per country. Fieldderived efficiencies — especially for informal processing, which this model does not separate at all — would materially change results for many countries.

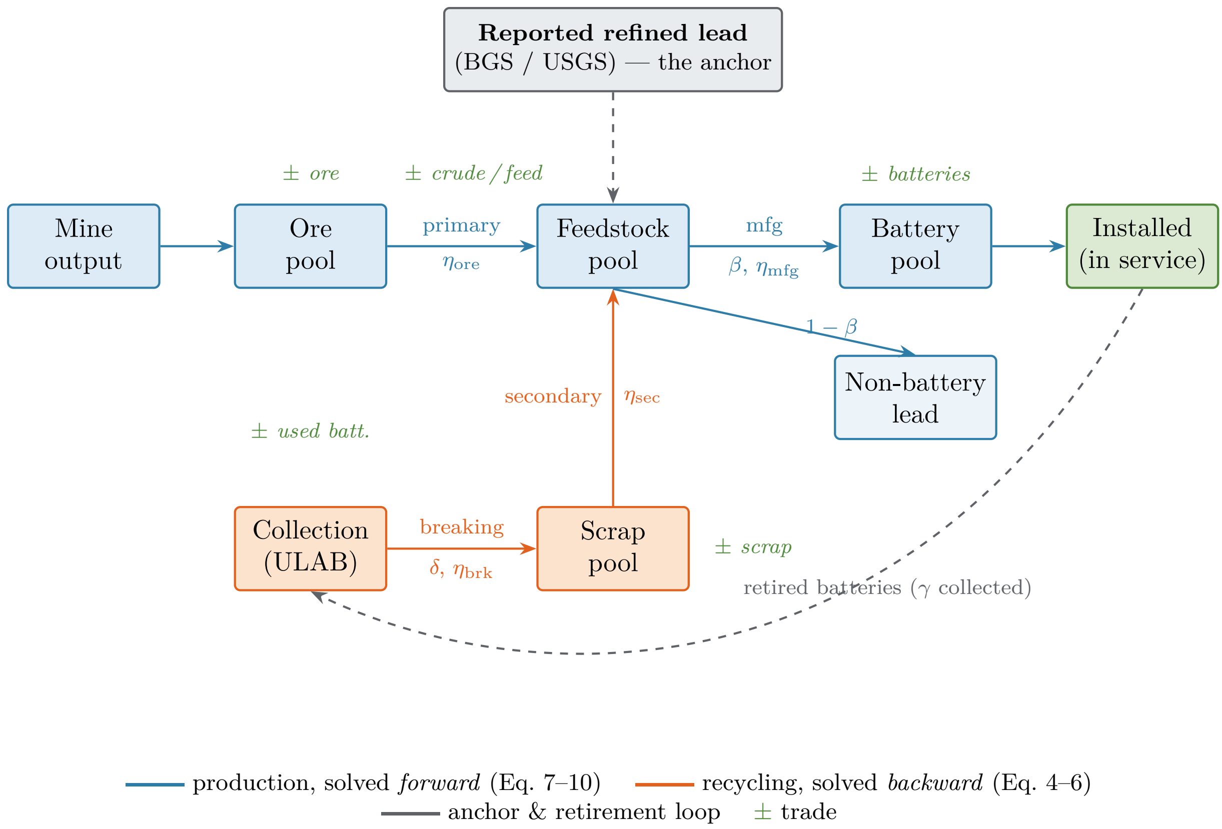
OPEN

Reported-data blind spots. Informal smelting, illicit trade, and irregular stockpiling are invisible to the model, and incomplete or provisional trade years pass straight through. **Small economies are least reliable**: at low volumes, a single mis-recorded shipment can dominate the picture. A multi-year average is preferred, and single-year figures should be read with caution.

6 References

- Dente, S.M.R., et al. (2025). *Assessing Lead Waste and Secondary Resources in Major Consumer Nations: A Vanishing Resource or a Toxic Legacy?* Resources, 14(4), 52. MDPI.
- Dong, D., et al. (2025). *Uncovering the anthropogenic lead cycle in China from 2000 to 2023: a dynamic material flow analysis.* Resources, Conservation & Recycling.
- Tür, M., Manhart, A., & Schleicher, T. (2016). *Generation of Used Lead-Acid Batteries in Africa — Estimating the Volumes.* Institute of Applied Ecology (Oeko-Institut), Freiburg.
- Data sources: BACI bilateral trade (CEPII); mine and refined-lead production, British Geological Survey (BGS) and US Geological Survey (USGS).

Figure 1. The material-flow loop and its anchor



The model pins the chain to reported refined-lead output (grey), then solves the recycling stages backward (orange) and the production stages forward (blue) from that anchor. Trade adjusts each pool (±). There is no time lag: the retired-battery loop (dashed) is treated as steady state within a single year or multi-year average.